

EXPERIMENT 10

240701426

a. First Fit

FACE Prep

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First Fit Contiguous Memory Allocation

Completed

Write a C program to implement the First Fit contiguous memory allocation technique. The program should take the sizes of various memory blocks and the sizes of processes as input. For each process, search through the memory blocks sequentially and allocate the first block that meets the size requirement. If no such block exists, indicate that the process cannot be allocated. Finally, display the allocation details and calculate internal fragmentation.

Input Format

- The first line contains an integer m representing the number of memory blocks.
- The next line contains m integers representing the sizes of each memory block.
- The next line contains an integer n representing the number of processes.
- The next line contains n integers representing the sizes of each process.

Output Format

- For each process, the program prints whether the process is allocated or not.
- If allocated, it displays the process number, process size, block

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main(){
4     int m,n;
5
6     printf("Enter number of memory blocks:");
7     scanf("%d",&m);
8
9     int blocksize[m],originalblock[m];
10
11     printf("\nEnter size of each block:\n");
12     for(int i=0;i<m;i++){
13         scanf("%d",&blocksize[i]);
14         originalblock[i]=blocksize[i];
15     }
16     printf("Enter number of processes:\n");
17     scanf("%d",&n);
18
19     int processsize[n];
20
21     printf("Enter size of each process:\n");
22     for(int i=0;i<n;i++){
23         scanf("%d",&processsize[i]);
24     }
25
26     for(int i=0;i<n;i++){
```

Test Cases 2

Run Sample Cases

Run All Cases

b. Best fit

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Best Fit Contiguous Memory Allocation

Completed

Write a C program to implement the Best Fit Contiguous Memory Allocation technique. The program should accept the sizes of memory blocks and the sizes of processes as input.

For each process, the program searches all available memory blocks and allocates the smallest block that is sufficient to satisfy the process size. If no suitable block is available, the process is marked as not allocated. The program should display the allocation details for each process and calculate the internal fragmentation.

Input Format

- The first line contains an integer representing the number of memory blocks.
- The second line contains the sizes of the memory blocks.
- The third line contains an integer representing the number of processes.
- The fourth line contains the sizes of the processes.

Output Format

- For each process, display whether it is allocated or not.
- If allocated, display the block number, block size, and internal fragmentation.

Select Language: C (GCC 9.2.0)

```
1 #include<stdio.h>
2 int main(){
3     int m,n;
4
5     printf("Enter number of memory blocks:\n");
6     scanf("%d",&m);
7
8     int blocksize[m],originalblock[m];
9
10    printf("Enter size of each block:\n");
11    for(int i=0;i<m;i++){
12        scanf("%d",&blocksize[i]);
13        originalblock[i] = blocksize[i];
14    }
15    printf("Enter number of processes:\n");
16    scanf("%d",&n);
17
18    int processsize[n];
19
20    printf("Enter size of each process:\n");
21    for(int i=0;i<n;i++){
22        scanf("%d",&processsize[i]);
23    }
24
25    for(int i=0;i<n;i++){
26        int bestindex = -1;
```

Test Cases 2

Run Sample Cases

Run All Cases